

EE 3326 Numerical and Scientific Data Analysis Using Python

Course Policies and Syllabus

Spring 2026

Ingram School of Engineering

Texas State University

Course:

Lecture: Friday 12:30 PM–3:20 PM IGRM 4103

Lecturer: Dr. Semih Aslan

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Course Description:

This course introduces Python programming for engineers. Topics include basics of Python programming, introduction to numerical Python (NumPy), scientific programming using Python (SciPy), data visualization using Matplotlib, data processing using Pandas and introduction to Object Oriented Programming using Python.

Prerequisites:

CS 1428 or CS 1342.

References:

- Paul J. Deitel, Harvey Deitel, *Python for Programmers: with Big Data and Artificial Intelligence Case Studies*, Pearson, 2019, ISBN-13:978-0135224335
- Sandeep Nagar, *Introduction to Python for Engineers and Scientists*, Apress, Berkeley, CA, 2018, ISBN-13:978-1-4842-3203-3
- Robert Johansson, *Numerical Python*, Apress, Berkeley, CA, 2019, ISBN-13:978-1-4842-4245-2
- Ossama Embarak, *Data Analysis and Visualization Using Python*, Apress, Berkeley, CA, 2018, ISBN-13:978-1-4842-4109-7
- Al Sweigart, *Automate the Boring Stuff with Python*, No Starch Press 2nd Ed., 2019, ISBN-13: 978-1593279929
- Mark Lutz, *Learning Python, 5th Edition*, O'Reilly Media, 2013, ISBN: 1449355730
- Luciano Ramalho, *Fluent Python: Clear, Concise, and Effective Programming*, 1st Edition, O'Reilly Media, 2015, ISBN: 1491946008
- Wes McKinney, *Python for Data Analysis: Data Wrangling with Pandas, NumPy, and I Python*, O'Reilly Media, 2nd Edition, 2017, ISBN: 1491957662
- Theodore Petrou, *Pandas Cookbook: Recipes for Scientific Computing, Time Series Analysis and Data Visualization using Python*, Packt Publishing, 2017, ISBN: 9781784393878

- Eric Matthes, *Python Crash Course*, No Starch Press 2nd Ed., 2019, ISBN-13: 978-1593279288

Course Objectives:

Upon completion of this course, the student will be able to understand and apply fundamental skills to learn:

- write Python programs
- use standard Python libraries
- use NumPy for numerical programming, vector, and matrix operations
- use SciPy for scientific programming
- demonstrate automated hardware verification using Python
- demonstrate data visualization using Matplotlib
- demonstrate data analysis using Pandas

Assessment of Student Learning: Assessment will be performed by the course instructor. The course grade will depend upon performance in the tests, projects, quizzes, and various assigned work. The final letter grade will be determined by the student's raw score as well as by the performance of the class (class average and standard deviation) as a whole. Additionally, attendance policies may have an impact on the letter grade.

Projects: Projects will be assigned on a regular schedule. Late Projects will not be accepted. I will drop one of your lowest project grades.

Exams: There will be two exams (Midterm and Final). These exams are scheduled as indicated in the course calendar. The final exam will be cumulative, but with an emphasis on the material covered since the midterm exam.

Quizzes: There will be quizzes during the semester. Missed quizzes may not be made up. We will drop one of your lowest Quiz grades.

Final Exam: 5/8/2026 2:00PM - 4:30PM

Grading:

Projects	30%
Midterm Exam	20%
Final Exam	25%
Final Project	10%
Quizzes	15%
Total	100%
A 90% – 100%	
B 80% – 89%	
C 70% – 79%	
D 60% – 69%	
F 0% – 59%	

Texas State University *Student Handbook* explains:

Grade A represents exceptionally high achievement as a result of aptitude, effort, and intellectual initiative.

Grade B represents a high achievement as a result of ability and effort.

Grade C represents average achievement.

Grade D represents the minimum passing grade.

Grade F represents unsatisfactory performance and indicates failure in the course.

Contribution to the Professional Component: This course develops professional skills and abilities in problems solving, programming, data analysis and system verification.

Working Together: It is ok to work together on projects. However, when it comes time for you to write up the solutions, I expect you to do this on your own, and it would be best for your own understanding if you put aside your notes from the discussions with your classmates and wrote up the solutions entirely from scratch. Working together on exams is not allowed.

Table 1: Tentative Course Schedule - University Academic Calender

Wk	Date	Topic	Projects	Quiz
1	Fr. 1/23	Introduction to Python and Python data types	xxx	xxx
2	Fr. 1/30	Python functions modules and packages	Pr.1	Q.1
3	Fr. 2/6	Python programming basics, decision making and loops	Pr.2	Q.2
4	Fr. 2/13	Introduction to numerical Python using NumPy	Pr.3	Q.3
5	Fr. 2/20	Introduction to numerical Python using NumPy	Pr.4	Q.4
6	Fr. 2/27	Advanced NumPy and Introduction to Matplotlib	Pr.5	Q.5
7	Fr. 3/6	Matplotlib	Pr.6	Q.6
8	Fr. 3/13	Midterm	Pr.7	xxx
9	Fr. 3/20	Spring Break	xxx	xxx
10	Fr. 3/27	Data analysis using Pandas	Pr.8	Q.7
11	Fr. 4/3	Data analysis using Pandas	Pr.9	Q.8
12	Fr. 4/10	Introduction to Object Oriented Programming using Python	F.Pr	Q.9
13	Fr. 4/17	Introduction to Object Oriented Programming using Python	F.Pr.	Q.10
14	Fr. 4/24	Introduction to Sympy	F.Pr	Q.11
15	Fr. 5/1	Final Review and Project Showcase	F.Pr	xxx
16	Fr. 5/8	Final Exam 2:00PM - 4.30PM	xxx	xxx

Computer: Students may use a computer during the exams. This will be explained during the exams.

Diversity, Equity, & Inclusion: In this course, we are committed to learning together to think for ourselves. Pursuit of this ideal relies upon civil behavior towards all course participants such that all have an equal opportunity for learning regardless of gender identity, gender expression, race, color, national or ethnic origin, religion or religious belief, political affiliation, age, marital status, sexual orientation, ability status, veteran status, familial background, or any other reason not related to intellectual merit. Participants in this course are expected to assist in sustaining this inclusive community.

Absences: You are expected to attend every class. Please read the exam, quiz, and project late policies carefully. An absence due to illness or family emergency may be excused, provided that you can supply acceptable written evidence if required, and that you notify your instructor *ASAP*. Notification is almost always possible immediately upon occurrence of an emergency. If you're too sick to telephone, you can get a friend to do it. Failure to make such timely notification may result in denial of your request.

Students with Disabilities: If you are a student with a disability who will require an accommodation(s) to participate in this course, please contact me as soon as possible. You will be asked to provide documentation from the Office of Disability Services. Failure to contact me in a timely manner may delay your accommodations. Also the Office of Distance and Extended Learning is committed to helping students with disabilities achieve their educational goals. Students who require special accommodations need to provide verification of their disability to the Office of Disability Services, Suite 5-5.1 LBJ Student Center, 512.245.3451.

Expectations: I expect that everyone will maintain a classroom conducive to learning. I like an informal atmosphere, but it must be orderly. Thus, everyone is expected to behave with basic politeness, civility, and respect for others. In particular, talking in class is ok if it's part of a class discussion or with me. Private communications are not, especially during quizzes and tests. Neither are reading extraneous materials, using electronic equipment, or sleeping. Please read the following Appendices from Student Handbook.